

assignment work

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:38pm

Quiz Instructions

there is a tutorial



Question 1 7 pts

A truck has four times the mass of a car and is moving with twice the speed of the car. If K_t and K_c refer to the kinetic energies of truck and car respectively, it is correct to say that.

Hint: the KE is proportional to mass and proportional to speed squared.

(mass is x4 and speed is x 2 but KE is proportional to speed squared.)

☐

$$K_t = K_c.$$

☐

$$K_t = 16K_c.$$

☐

$$K_t = \frac{1}{2} K_c.$$

☐

$$K_t = 2K_c.$$

☐

$$K_t = 4K_c.$$



Question 2 7 pts

A child does 350 J of work while pulling a box from the ground up to his tree house at a steady speed with a light rope. The tree house is 4.0 m above the ground. What is the mass of the box? hints:

hint: The 2 forces acting on the box are the child pull and its weight. If the speed is constant, the pull=weight of the box. Work done by child = his pull x distance.

☐

6.7 kg

☐

8.0 kg

☐

5.3 kg



8.9 kg



Question 3 7 pts

You slam on the brakes of your car in a panic, and skid a certain distance on a straight level road. If you had been traveling twice as fast, what distance would the car have skidded, under the same conditions? hint: Remember the KE is proportional to speed squared. That KE loss turns to force x distance (force is the same)



It would have skidded one half as far.



It would have skidded 4 times farther.



It would have skidded 1.4 times farther.



It is impossible to tell from the information given.



It would have skidded twice as far.



Question 4 7 pts

Which one has larger kinetic energy: a 500-kg object moving at 40 m/s or a 1000-kg object moving at 20 m/s?

hint: KE is proportional to mass and inversely proportional to speed squared. Here mass is x 2 and speed is divided by 2. Or just compute KE.



the 1000-kg object



the 500-kg object



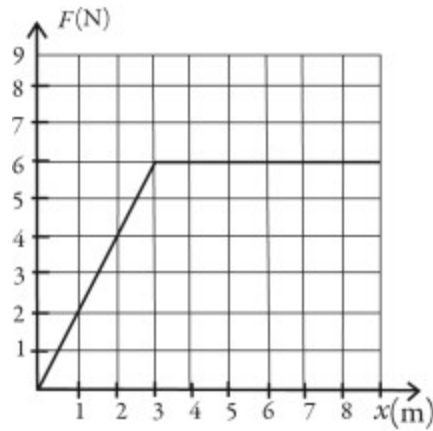
Both have the same kinetic energy.



Question 5 7 pts

A graph of the force on an object as a function of its position is shown in the figure. Determine the amount of work done by this force on an object that moves from $x = 1.0$ m to $x = 6.0$ m. (Assume an accuracy of 2 significant figures for the numbers on the graph.)

hint: work done is the area underneath the curve. You see a triangle and a square it seems.


☐

35 J

☐

22 J

☐

29 J

☐

26 J

☐

27 J

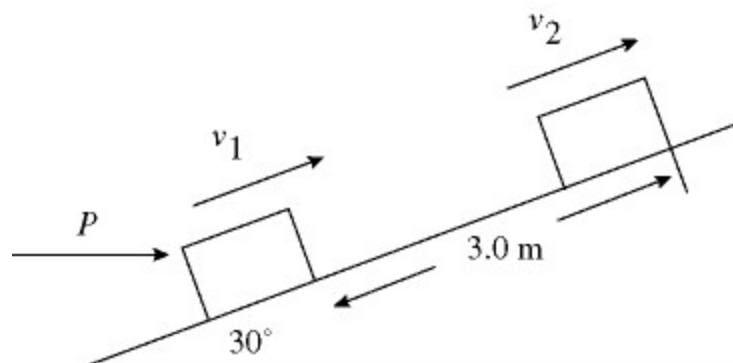
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Question 6 7 pts

In the figure, a 700-kg crate is on a rough surface inclined at 30° . A constant external force $P = 5600$ N is applied horizontally to the crate. As the force pushes the crate a distance of 3.00 m up the incline, the speed changes from 1.40 m/s to 2.50 m/s.

How much work does gravity do on the crate during this process? (about)

hint: tricky question but interesting. As usual, take the x-axis along the incline. The work done by gravity is the component of the weight along the x-axis multiplied by the displacement. The work is negative as it opposes the motion. !!!! you can not find the work done by gravity by subtracting work done by P from change in KE. It says the surface is rough. So we don't know the work done by friction. tricky!! read carefully.



☐

-3400 J

☐

+10,300 J

☐

zero

☐

+3400 J

☐

-10,300 J

☐

Question 7 7 pts

Determine the scalar product of $\vec{A} = 6.0\hat{i} + 4.0\hat{j} - 2.0\hat{k}$ and $\vec{B} = 5.0\hat{i} - 6.0\hat{j} - 3.0\hat{k}$.

☐
 $30\hat{i} - 24\hat{j} + 6\hat{k}$
☐
 $30\hat{i} + 24\hat{j} + 6\hat{k}$
☐

12

☐

60

☐

undefined

☐

Question 8 7 pts

Determine the angle between the directions of vector $\vec{A} = 3.00\hat{i} + 1.00\hat{j}$ and vector $\vec{B} = -3.00\hat{i} + 3.00\hat{j}$.

☐

45.2°

☐

30.0°

☐

88.1°

☐

117°

☐

26.6°

☐

Question 9 7 pts

A 1.0-kg object moving in a certain direction has a kinetic energy of 2.0 J. It hits a wall and comes back with half its original speed. What is the kinetic energy of this object at this point?

☐

4.0 J



2.0 J



0.25 J



0.50 J



1.0 J



Question 10 7 pts

A 1000-kg car experiences a net force of 9500 N while slowing down from 30 m/s to 15.9 m/s. How far does it travel while slowing down?

hint: apply the work-KE theorem. change in KE = Work done by force



31 m



41 m



37 m



34 m



Question 11 7 pts

A 10-kg dog is running with a speed of 5.0 m/s. What is the minimum work required to stop the dog in 2.40 s?

hint: Just apply work-KE theorem



100 J



75 J



125 J



50 J



Question 12 7 pts

Find the net work done by friction on the body of a snake slithering in a complete circle of 3.93 m radius. The coefficient of friction between the ground and the snake is 0.25, and the snake's weight is 54.0 N.

hint: its not zero.



0 J



-670 J



-3300 J



-330 J



Question 13 7 pts

A crane lifts a 425 kg steel beam vertically a distance of 117 m. How much work does the crane do on the beam if the beam accelerates upward at 1.8 m/s^2 ? Neglect frictional forces.

hint: !!! you are looking for the work done by the crane. Find the tension first ($F_{\text{net}} = ma$) , Work = T x displacement.

 $4.0 \times 10^5 \text{ J}$  $5.8 \times 10^5 \text{ J}$  $4.9 \times 10^5 \text{ J}$  $3.4 \times 10^5 \text{ J}$ 

Question 14 7 pts

An airplane flies 120 km at a constant altitude in a direction 30.0° north of east. A wind is blowing that results in a net horizontal force on the plane due to the air of 2.40 kN in a direction 10.0° south of west. How much work is done on the plane by the air?

hint: you have to draw. Find the component of the force along the displacement.

 $0.821 \times 10^8 \text{ J}$  $-221 \times 10^8 \text{ J}$  $-2.71 \times 10^8 \text{ J}$  $-0.985 \times 10^8 \text{ J}$  $221 \times 10^8 \text{ J}$ 

Question 15 7 pts

In the figure, a constant external force $P = 160 \text{ N}$ is applied to a 20.0-kg box, which is on a rough horizontal surface. While the force pushes the box a distance of 8.00 m , the speed changes from 0.500 m/s to 2.60 m/s . The work done by friction during this process is closest to

hint: $F_{\text{net}} = P_x - \text{friction}$. And $F_{\text{net}} \times \text{displacement} = \text{change in KE}$.



☐ +1170 J.

☐ +1040 J.

☐ -1040 J.

☐ +1110 J.

☐ -1170 J.



Question 16 5 pts

A person is pulling a sled. The work done by the normal force on the sled is

(bonus)

☐ IDK

☐ negative

☐ positive

☐ 0

Not saved

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